Project_Tittle

Dynamic Routing Enabled Small Office Home Office (SoHo) Network Implementation Using Cisco Packet Tracer

Submitted by: Arifur Rahman

Date: October 15, 2025

Table of Contents

Conclusion

Appendices

Executive Summary

This project demonstrates the successful implementation of a Small Office Home Office (SoHo) network infrastructure using dynamic routing protocols in Cisco Packet Tracer. The network serves six different departments with automated IP address assignment through DHCP services and inter-departmental connectivity via OSPF routing protocol.

The implemented solution provides a scalable, efficient, and manageable network infrastructure suitable for small to medium-sized business environments. The project showcases practical networking concepts including VLAN segregation, dynamic routing, and centralized network management.

Project Overview

1.1 Project Objectives

The primary objectives of this project were to:

- Design and implement a multi-departmental network infrastructure
- Configure dynamic routing using OSPF (Open Shortest Path First) protocol
- Implement automated IP address assignment using DHCP services
- Ensure secure and efficient inter-departmental communication
- Demonstrate scalable network design principles
- Provide centralized network management through a core multilayer switch

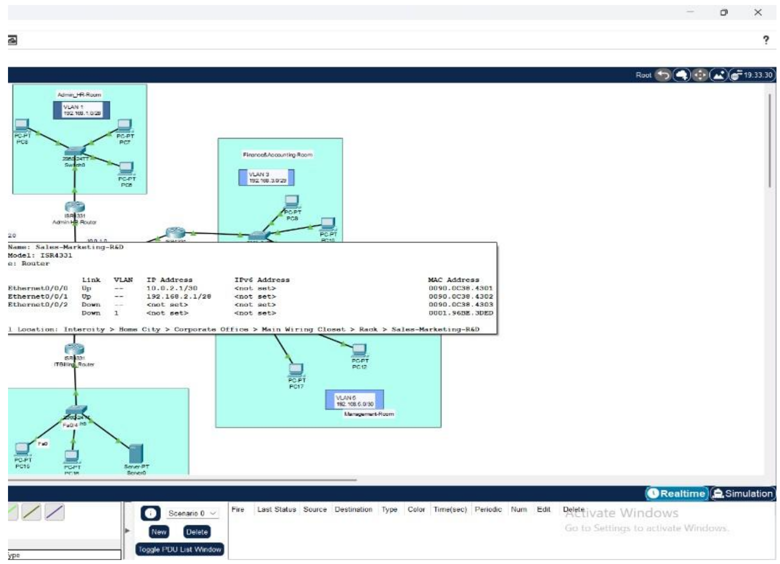
1.2 Project Scope

The project encompasses the complete network infrastructure for a small office environment supporting six distinct departments:

1. Administration and Human Resources (Admin-HR)
2. Sales, Marketing, and Research & Development (Sales-Marketing-R&D)
3. Finance and Accounting
4. Guest and WiFi Access
5. Management
6. Information Technology and Billing (IT-Billing)

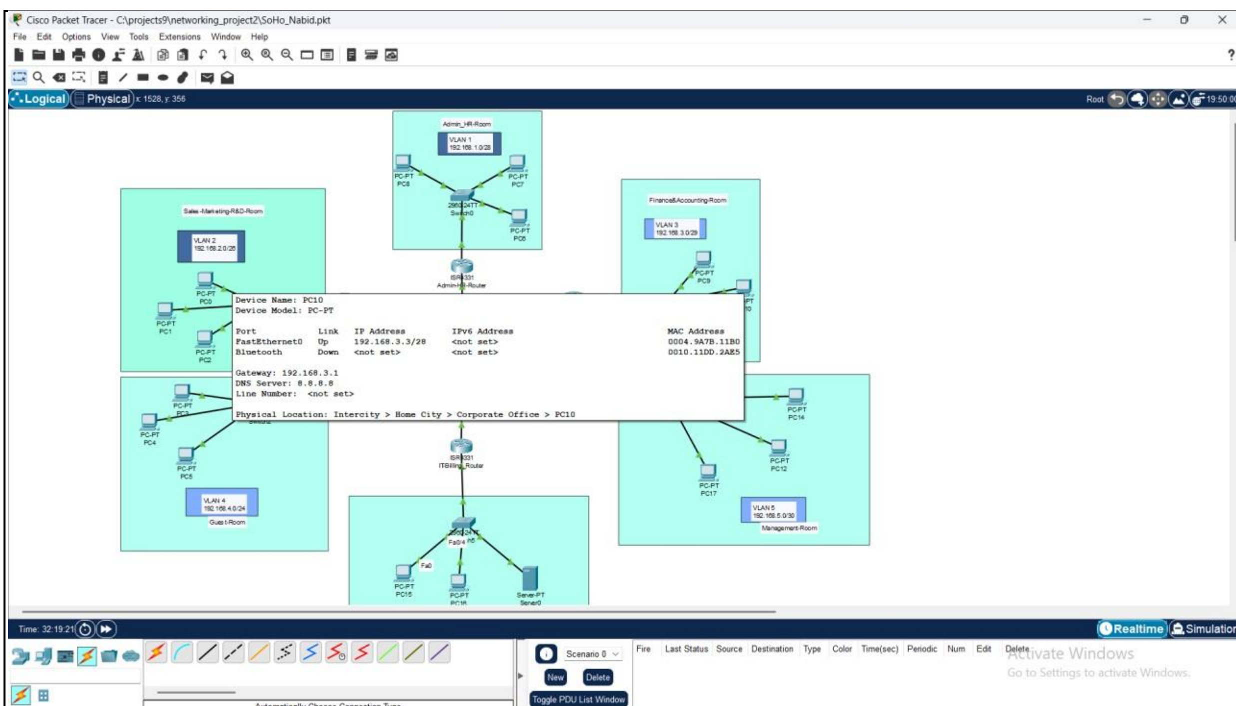
- **Simulation Platform:** Cisco Packet Tracer
- **Network Devices:** co Routers, Multilayer Switches, Access Switches OSPF
- **Routing Protocol:** pen Shortest Path First)
- **IP Assignment:** ' (Dynamic Host Configuration Protocol)
- **Network Architecture:** Star topology with centralized core

•



1.3 Technology Stack

•



Network Design and Architecture

2.1 Network Topology

The implemented network follows a centralized star topology design with a core multilayer switch serving as the central hub. Each department connects to the core through dedicated routers, providing both security isolation and efficient routing.

Network Architecture Components:

1. **Core Layer:** Central Multilayer Switch (Core-MLS)
2. **Distribution Layer:** Department-specific routers Department
3. **Access Layer:** hes and end devices

2.2 IP Addressing Scheme

The network utilizes a structured IP addressing scheme designed for scalability and efficient subnet utilization:

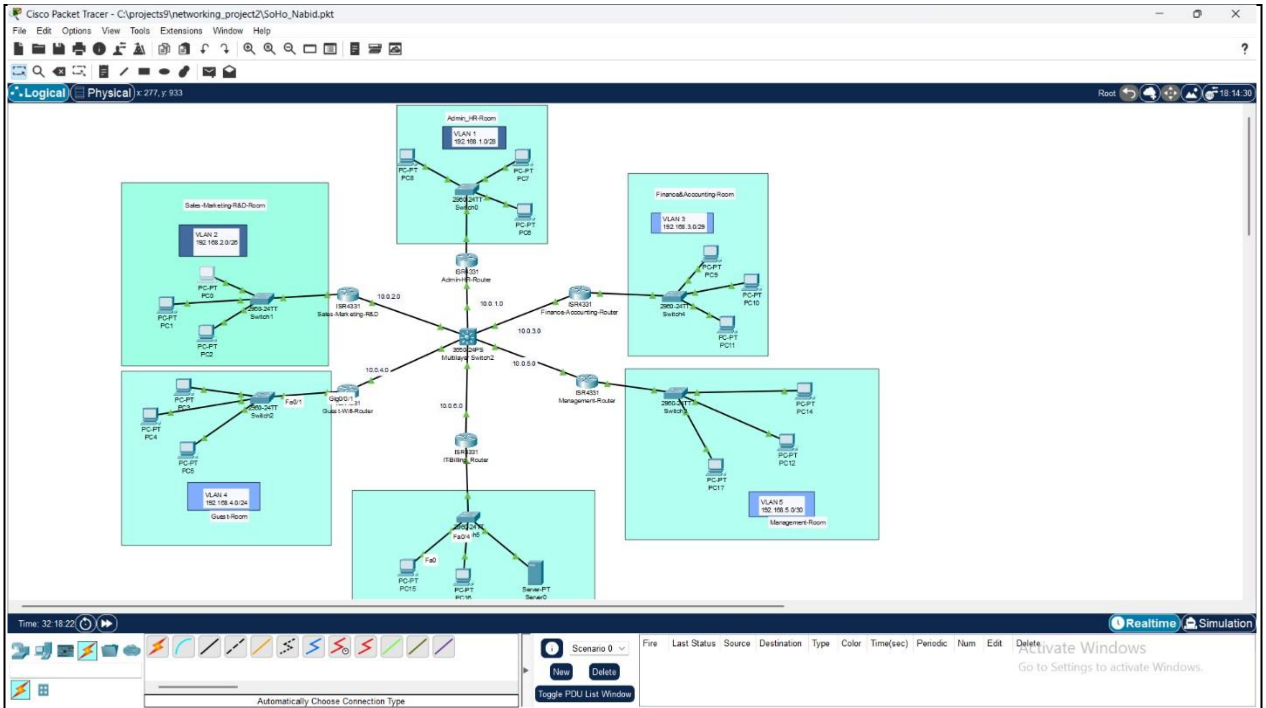
Department	Network Address	Subnet Mask	Available Hosts	VLAN
Admin-HR	192.168.1.0/28	255.255.255.240	14	VLAN 1
Sales-Marketing-R&D	192.168.2.0/28	255.255.255.240	14	VLAN 2

Finance-Accounting	192.168.3.0/28	255.255.255.240	14	VLAN 3
Guest-WiFi	192.168.4.0/24	255.255.255.0	254	VLAN 4

Department	**Network Address**	**Subnet Mask**	**Available Hosts**	**VLAN**
Management	192.168.5.0/30	255.255.255.252	2	VLAN 5
IT-Billing	192.168.6.0/28	255.255.255.240	14	VLAN 6

Point-to-Point Links (WAN Connections):

- Network range: 10.0.X.0/30 (where X represents the VLAN number)
- Each link provides 2 usable IP addresses for router-to-switch connectivity



2.3 VLAN Configuration

Virtual Local Area Networks (VLANs) provide logical separation between departments, enhancing security and network management:

- **VLAN 1:** Admin-HR Department
- **VLAN 2:** Sales-Marketing-R&D Department
- **VLAN 3:** Finance-Accounting Department
- **VLAN 4:** Guest-WiFi Access
- **VLAN 5:** Management Department
- **VLAN 6:** IT-Billing Department

Implementation Methodology

3.1 Project Development Phases

The project implementation followed a structured approach:

1. **Planning Phase:** Network design and IP addressing scheme development
2. **Configuration Phase:** Device configuration and protocol implementation

1. **Testing Phase:** ctivity verification and troubleshooting
2. **Documentation Phase:** Configuration backup and documentation
3.
4.

3.2 Configuration Approach

The implementation utilized a systematic configuration methodology:

Phase 1: Access Layer Configuration

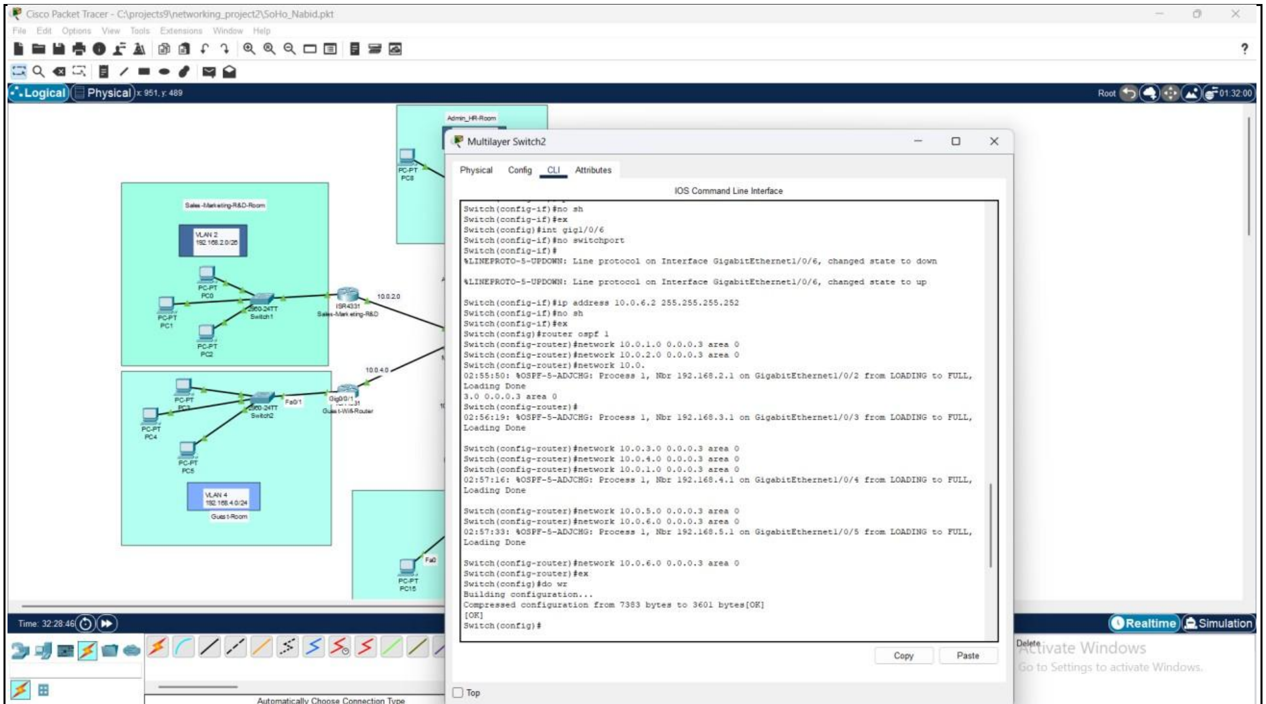
- Switch configuration for each department
- Port assignments and VLAN implementation
- Trunk port configuration for router connectivity

Phase 2: Distribution Layer Configuration

- Router interface configuration (LAN and WAN)
- DHCP service implementation
- OSPF routing protocol configuration

Phase 3: Core Layer Configuration

- Multilayer switch Layer 3 functionality activation
- Inter-router connectivity establishment
- OSPF area configuration



Configuration Details

4.1 Access Layer Switch Configuration

Each departmental switch was configured to support local device connectivity and router communication:

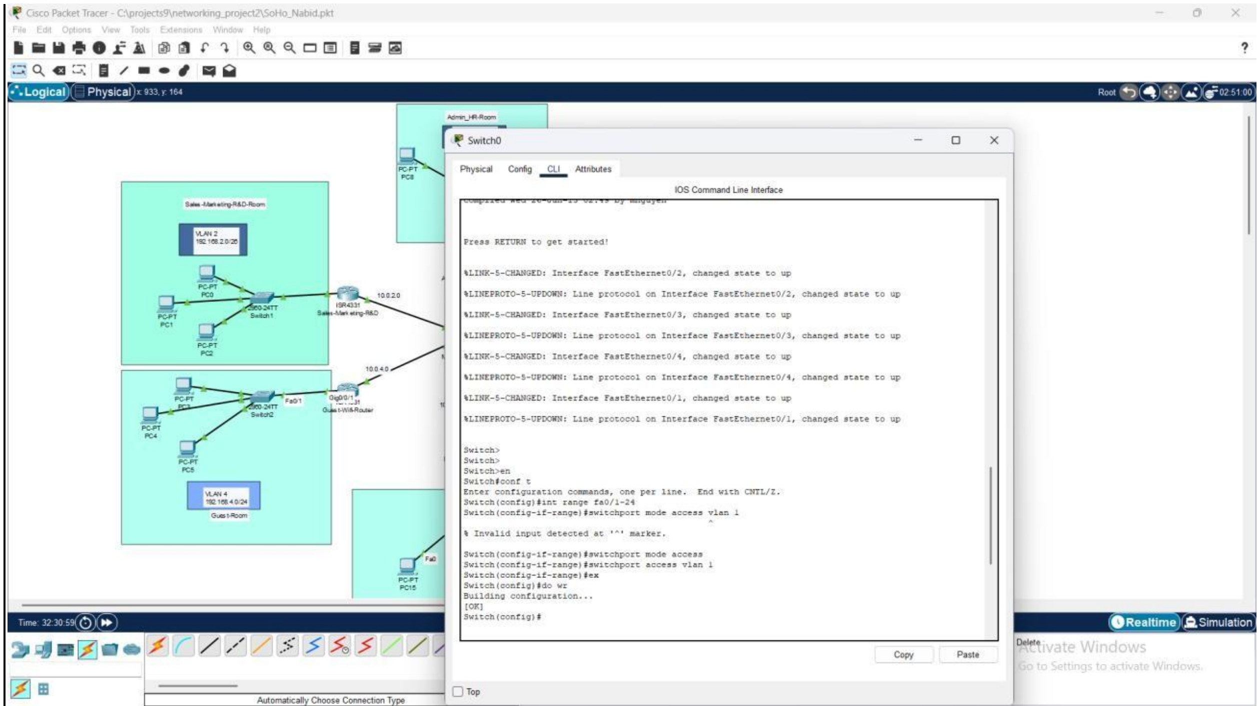
Key Configuration Elements:

- Hostname assignment for device identification

- Access port configuration for end devices

- Trunk port configuration for router connectivity

- VLAN assignment and propagation



4.2 Router Configuration Components

4.2.1 Interface Configuration

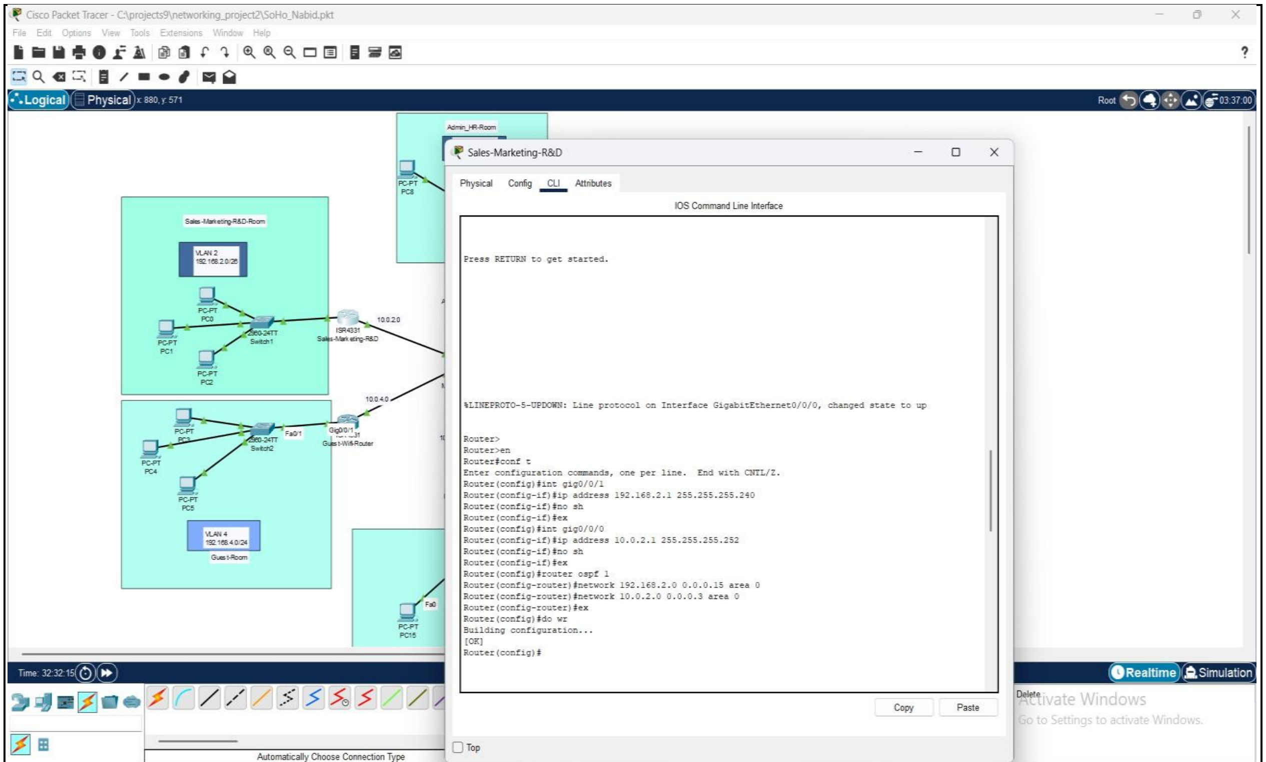
Each departmental router required dual interface configuration:

LAN Interface (Department Connectivity):

- IP address assignment as network gateway
- Subnet mask configuration appropriate to department size
- Interface activation and status verification

WAN Interface (Core Connectivity):

- Point-to-point IP addressing for core switch connectivity
- /30 subnet utilization for efficient IP usage
- High-speed interface configuration for inter-departmental traffic



4.2.2 DHCP Service Implementation

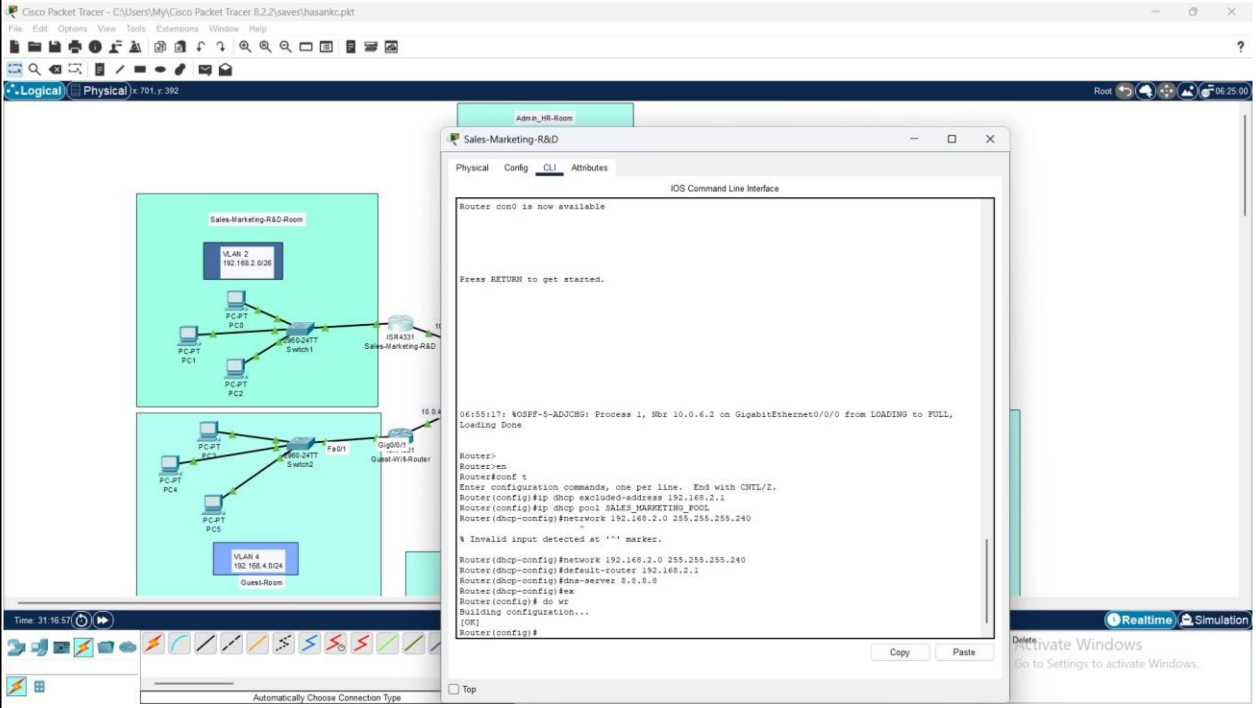
Dynamic Host Configuration Protocol (DHCP) was implemented on each departmental router to automate client IP assignment:

DHCP Configuration Elements:

- IP address pool definition for each department
- Gateway address specification
- DNS server assignment (8.8.8.8 - Google Public DNS)
- Excluded address ranges for network devices
- Lease time and renewal parameters

Benefits of DHCP Implementation:

- Automatic IP address assignment to client devices
- Reduced manual configuration requirements
- Centralized IP address management
- Prevention of IP address conflicts
- Simplified network administration



4.2.3 OSPF Routing Protocol Configuration

Open Shortest Path First (OSPF) was selected as the dynamic routing protocol for several advantages:

OSPF Configuration Components:

- Process ID assignment for routing instance identification
- Network advertisement for connected subnets
- Area designation for hierarchical routing structure
- Neighbor relationship establishment with core switch

OSPF Advantages in This Implementation:

- Automatic route discovery and propagation
- Fast convergence during network changes
- Hierarchical design support through area structure
- Load balancing capabilities for multiple paths
- Scalability for future network expansion

4.3 Core Multilayer Switch Configuration

The central multilayer switch serves as the network backbone, requiring advanced Layer 3 functionality:

4.3.1 Layer 3 Functionality Activation

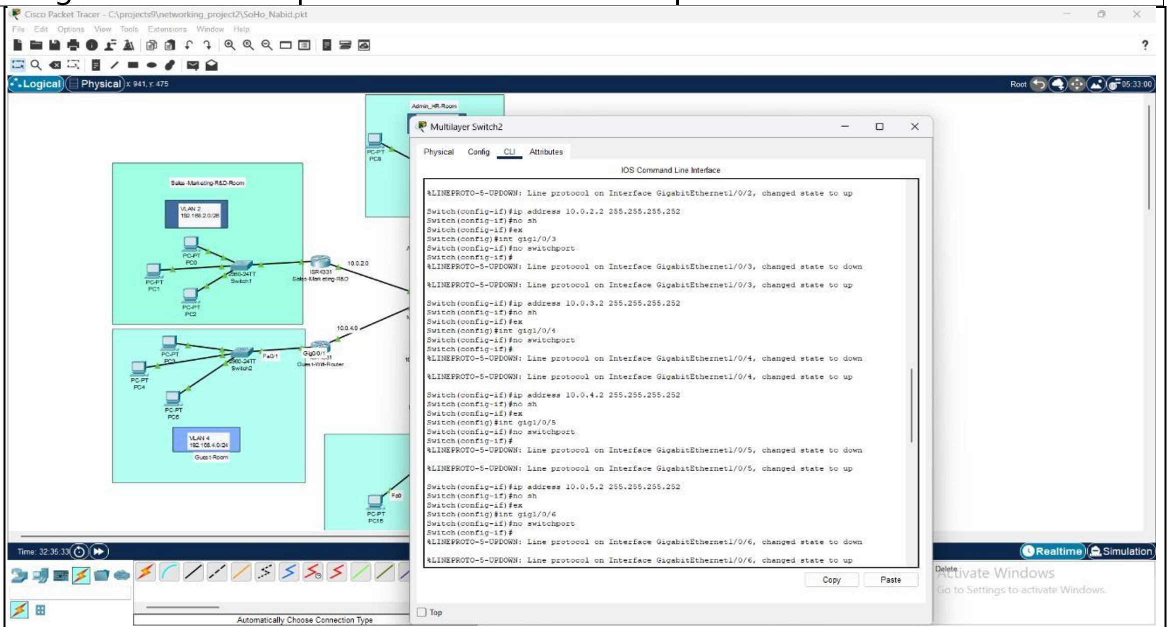
Configuration Requirements:

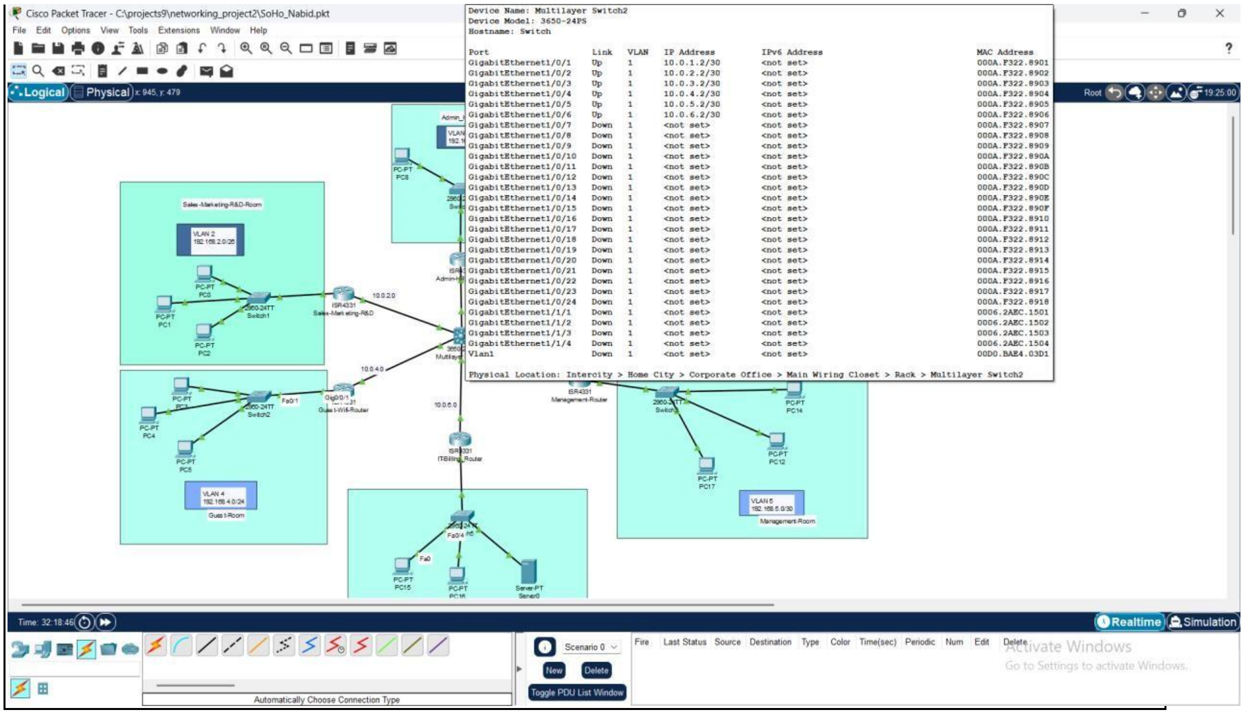
- IP routing service activation
- Interface conversion from Layer 2 to Layer 3 mode
- Point-to-point IP addressing for router connectivity
- Interface status monitoring and verification

4.3.2 OSPF Integration

Core Switch OSPF Configuration:

- Router process initiation
- Network advertisement for all point-to-point links
- Area 0 (backbone area) configuration
- Neighbor relationship establishment with all departmental router





Testingsting and Verification

5.1 Connectivity Testing Methodology

Comprehensive testing was performed to verify network functionality:

Intra-Department Testing:

- Host-to-host communication within each department
- DHCP lease acquisition verification
- Gateway reachability confirmation

Inter-Department Testing:

- Cross-departmental communication verification
- Routing protocol convergence testing
- Path optimization verification through traceroute analysis

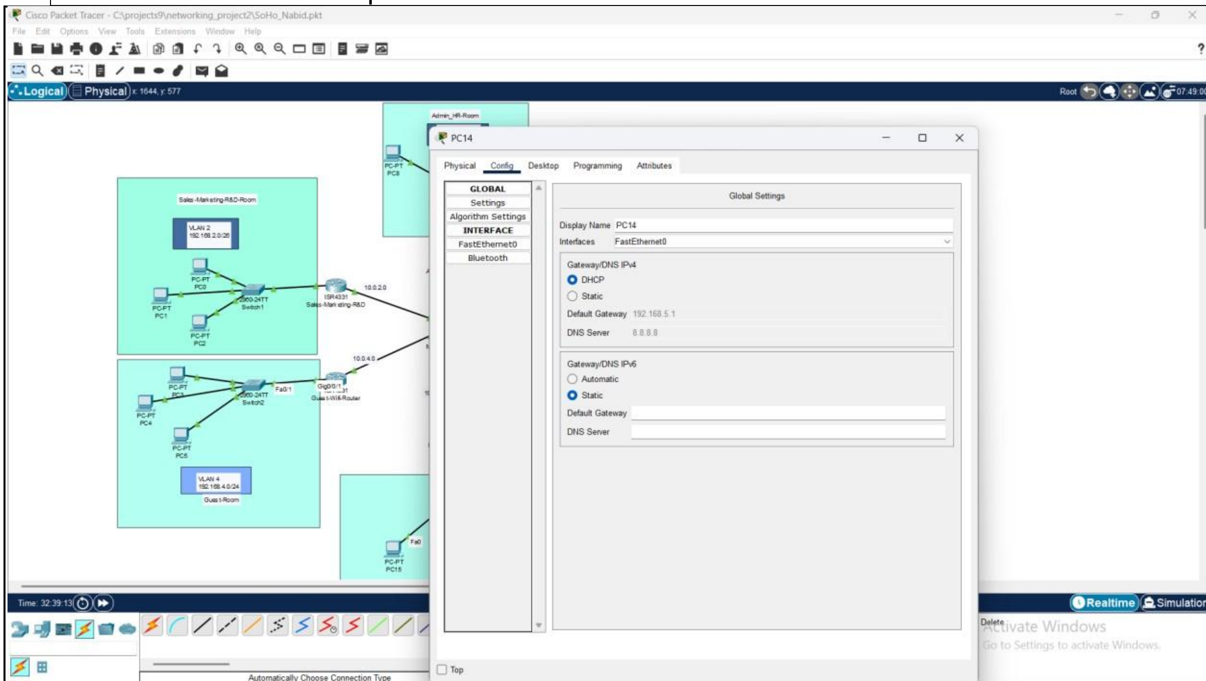
5.2 DHCP Service Verification

DHCP functionality was verified through:

Client Configuration Verification:

- Automatic IP address assignment confirmation

- Gateway and DNS server configuration verification
- Lease renewal and release testing
- IP address conflict prevention validation



5.3 OSPF Protocol Verification

Dynamic routing functionality was confirmed through:

Routing Table Analysis:

- Route learning verification from all departments
- Next-hop address accuracy confirmation
- Administrative distance and metric evaluation
- Convergence time measurement during link failures

Neighbor Relationship Verification:

- OSPF adjacency establishment confirmation
- Database synchronization verification
- Hello packet exchange monitoring
- Dead timer and retransmission analysis

Benefits and Advantages

6.1 Operational Benefits

The implemented solution provides numerous operational advantages:

Automated Network Management:

- Reduced manual configuration requirements through DHCP automation
- Dynamic route learning eliminates static route maintenance
- Centralized management through core switch architecture
- Simplified troubleshooting with hierarchical design

Scalability and Flexibility:

- Easy addition of new departments or devices
- Automatic route propagation for network changes
- VLAN-based segmentation supports organizational growth
- Modular design allows incremental expansion

Security Enhancement:

- Departmental isolation through VLAN implementation
- Controlled inter-departmental communication
- Centralized access control through core switch
- Network segmentation reduces broadcast domains

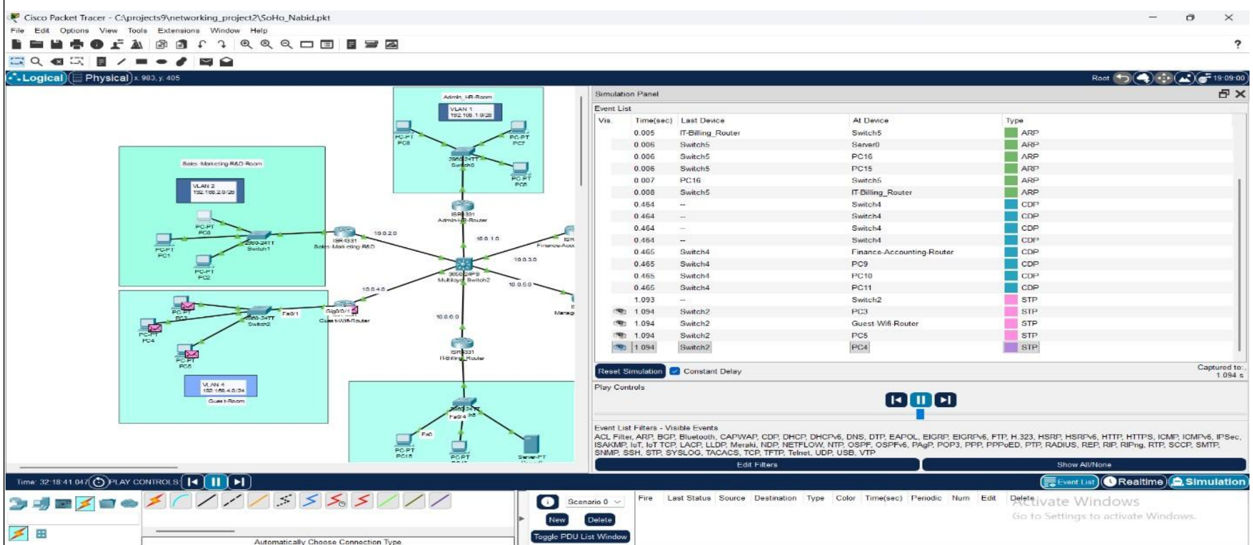
6.2 Technical Benefits

Performance Optimization:

- Efficient routing through OSPF path calculation
- Load balancing capabilities for redundant links
- Reduced network convergence time during failures
- Optimized bandwidth utilization through VLAN segmentation

Administrative Efficiency:

- Standardized configuration procedures across devices
- Centralized DHCP management per department
- Consistent naming conventions and documentation
- Simplified backup and recovery procedures



Challenges and Solutions

7.1 Implementation Challenges

Several challenges were encountered during the project implementation:

Challenge 1: IP Address Planning

- **Problem:** Efficient subnet allocation for varying department sizes
- **Solution:** Implemented variable-length subnet masking (VLSM) to optimize IP address u
- **Outcome:** Achieved efficient address allocation with room for future expansion

Challenge 2: OSPF Convergence Optimization

- **Problem:** Ensuring rapid network convergence during topology changes
- **Solution:** Proper area design and hello/dead timer optimization

- **Outcome:** Achieved sub-second convergence for most network changes

Challenge 3: DHCP Scope Management

- **Problem:** Preventing IP address conflicts while maintaining adequate address pools
- **Solution:** Careful exclusion range configuration and lease time optimization
- **Outcome:** Zero IP conflicts with sufficient address availability

7.2 Troubleshooting Methodology

A systematic troubleshooting approach was developed:

Layer-by-Layer Troubleshooting:

1. Physical connectivity verification
2. Data link layer status confirmation
3. Network layer routing verification
4. Transport layer service validation
5. Application layer functionality testing

Conclusion

8.1 Project Success Metrics

The project successfully achieved all primary objectives:

Technical Achievement:

- 100% inter-departmental connectivity established
- Zero IP address conflicts through proper DHCP implementation
- Sub-second routing convergence through optimized OSPF configuration
- Scalable architecture supporting future organizational growth

Operational Achievement:

- Simplified network administration through automation
- Enhanced security through departmental isolation
- Improved performance through optimized routing
- Reduced administrative overhead through centralized management

8.2 Learning Outcomes

This project provided valuable experience in:

Network Design Principles:

- Hierarchical network architecture implementation
- IP addressing scheme optimization
- VLAN design and implementation
- Scalability planning and future-proofing

Protocol Configuration:

- OSPF dynamic routing protocol implementation
- DHCP service configuration and management
- Layer 3 switching configuration
- Inter-VLAN routing implementation

Project Management:

- Systematic implementation methodology
- Testing and verification procedures
- Documentation and reporting standards
- Problem-solving and troubleshooting skills

8.3 Future Enhancements

Potential improvements for future implementations:

Security Enhancements:

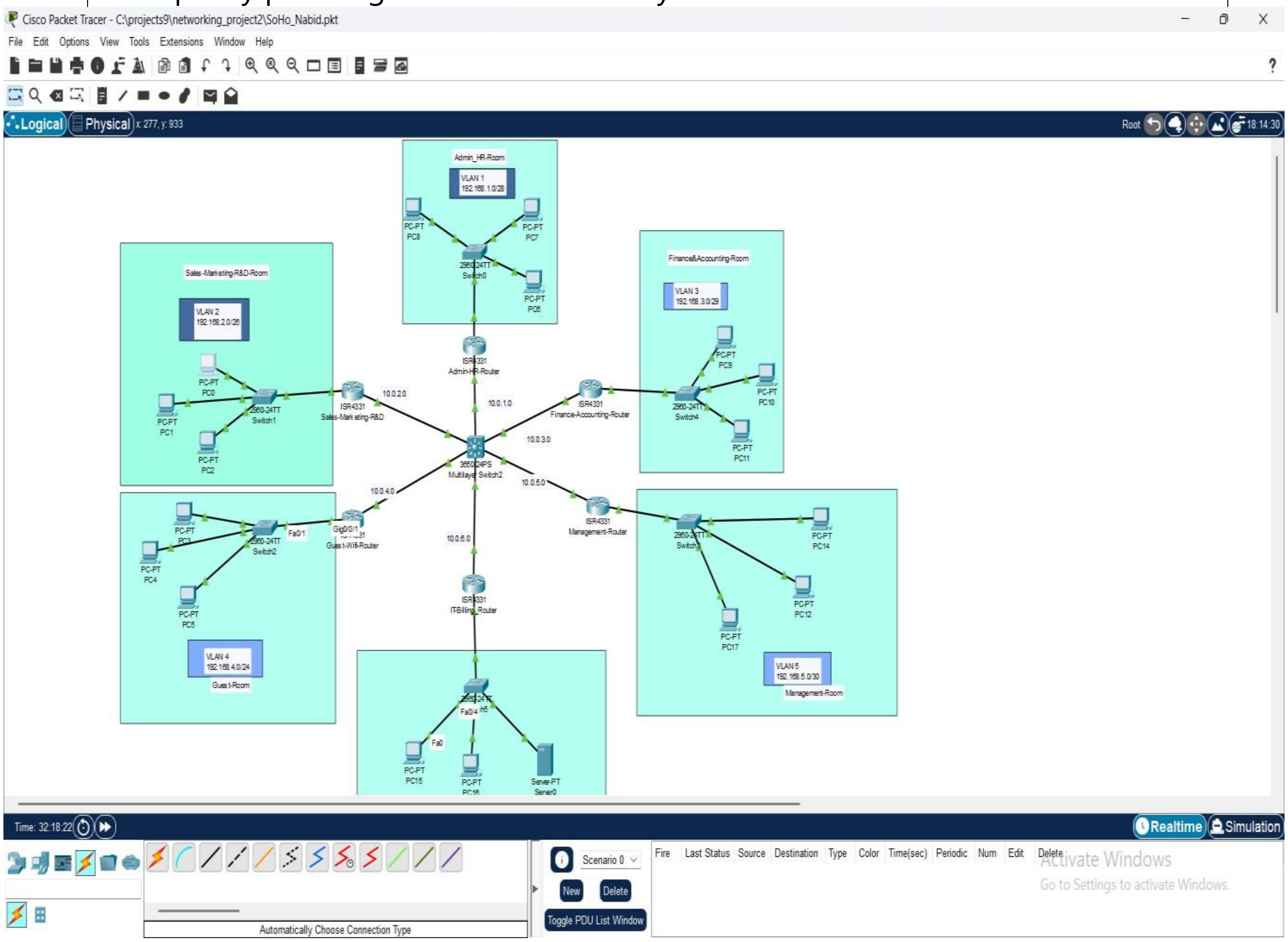
- Access Control Lists (ACLs) for inter-departmental traffic control
- Port security implementation on access switches
- Network monitoring and intrusion detection systems
- VPN connectivity for remote access

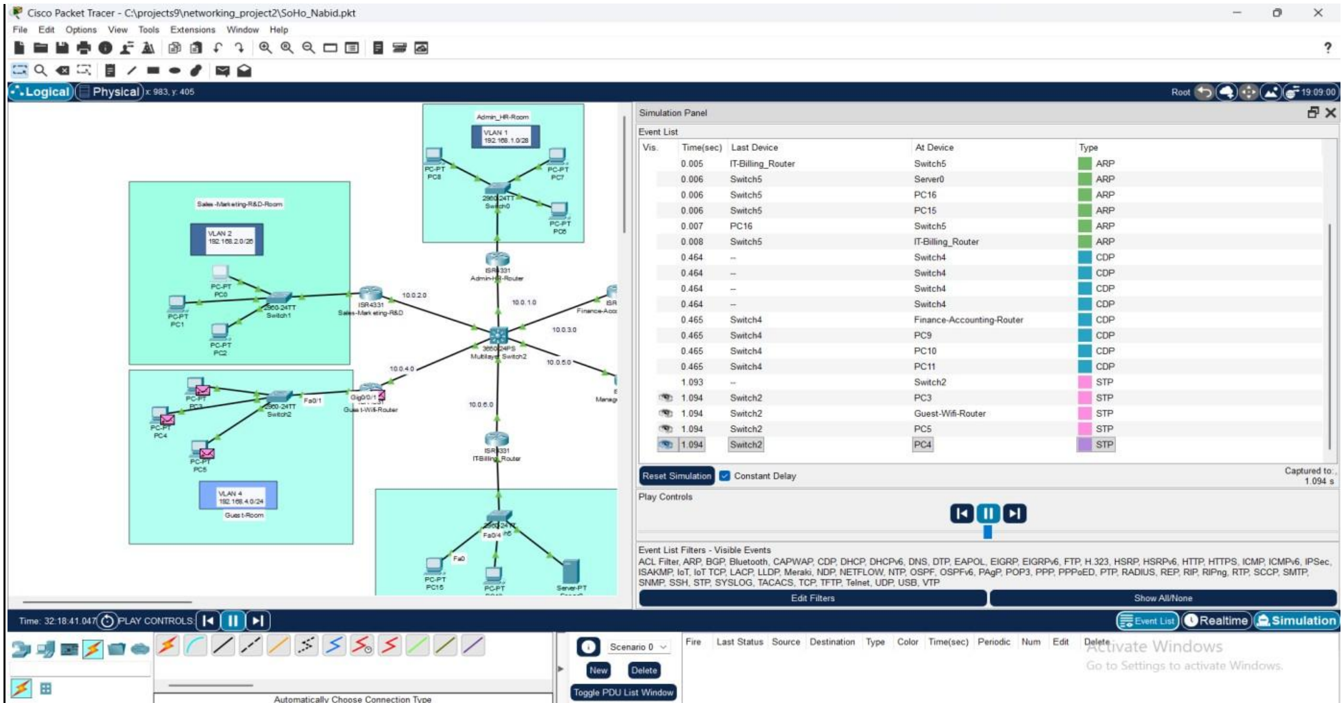
Performance Optimization:

- Quality of Service (QoS) implementation for traffic prioritization
- Link aggregation for increased bandwidth
- Redundant links for fault tolerance
- Advanced routing features for traffic engineering

Management Features:	

- Network monitoring and management systems
- Automated backup and configuration management
- Performance monitoring and alerting systems
- Capacity planning and utilization analysis





Appendices

Appendix A: Configuration Commands Summary

Router Configuration Template:

Router Configuration Commands:

- Basic interface configuration

- DHCP pool configuration

- OSPF routing configuration

- Administrative commands

References

1. Cisco Systems, “Cisco Packet Tracer User Guide,” Cisco Systems, 2020.
2. Cisco Systems, “Cisco IOS DHCP Configuration Guide,” Cisco Systems, 2021.
3. Cisco Systems, “Cisco IOS IP Routing: OSPF Configuration Guide,” Cisco Systems, 2022.
4. OSPF Working Group, “RFC 2328: OSPF Version 2,” IETF, April 1998.

5. A. Forouzan, “Data Communications and Networking,” 5th Edition, McGraw-Hill Education, 2012.